

Zadanie 26

Oblicz granicę $\lim_{n \rightarrow \infty} \frac{\sqrt{9n^2 - 2n} - 3n}{\sqrt{n+5} - 2}$.

① Usuwamy niewymierność.

$$\lim_{n \rightarrow \infty} \frac{\sqrt{9n^2 - 2n} - 3n}{\sqrt{n+5} - 2} \cdot \frac{\sqrt{9n^2 - 2n} + 3n}{\sqrt{9n^2 - 2n} + 3n} = \lim_{n \rightarrow \infty} \frac{9n^2 - 2n - 9n^2}{(\sqrt{n+5} - 2)(\sqrt{9n^2 - 2n} + 3n)} =$$

② Obliczamy granicę.

$$\lim_{n \rightarrow \infty} \frac{-2n}{\sqrt{n} \left(\sqrt{1 + \frac{5}{n}} - \frac{2}{\sqrt{n}} \right) \left(\sqrt{9 - \frac{2}{n}} + 3 \right)} = \lim_{n \rightarrow \infty} \frac{-2}{\sqrt{n} \cdot 1 \cdot (3+3)} = \lim_{n \rightarrow \infty} \frac{\cancel{\sqrt{n}} \cdot \left(\frac{-2}{\sqrt{n}} \right)^{\rightarrow 0}}{\cancel{\sqrt{n}} \cdot 6} = 0$$

Odp.: $\lim_{n \rightarrow \infty} \frac{\sqrt{9n^2 - 2n} - 3n}{\sqrt{n+5} - 2} = 0.$